Noisy Channel Inference in Large Language Models

Background: Human communication occurs in the presence of noise (e.g., speaker errors). Comprehenders are thought to interpret the meaning of a sentence by integrating priors (world/language knowledge) and a noise likelihood [1, 2]. Implausible sentences (e.g., "The mother gave the candle the daughter") which may have resulted from a deletion elicit more non-literal interpretations — inferences — than those which may have resulted from an insertion (e.g., "The mother gave the daughter to the candle"), reflecting that deletions constitute more probable noise. Both elicit more inferences in supportive contexts, where the prior probability of a more plausible alternative is higher, relative to unsupportive contexts [3]. Prior work shows that LLM responses are influenced by prior probability [4], and provides preliminary evidence that ChatGPT interpreted implausible sentences non-literally [5]. Here, we systematically test whether LLMs engage in noisy-channel inference, probing whether their priors and noise likelihoods are calibrated to human behavior.

Methods: We used four OpenAI models (GPT-4o, GPT-4.1, GPT-o3, GPT-5), and one open-source model (LLama 3.3 70B-Instruct). All models were evaluated in English using a fixed system prompt ("You are a helpful assistant. You will get stories and you have to answer yes or no to a question about the story. Only provide yes or no without any punctuation or additional detail.") and a user prompt. Each user prompt (trial) consisted of two context sentences, a target sentence, and a comprehension question as in [3] (**Table 1**). Every run (3 runs per model) contained 20 target trials (10 plausible, 10 implausible), and 48 filler trials. Target sentences varied by structure: double-object (DO; one deletion) vs. prepositional-object (PO; one insertion) or active vs. passive (two-word exchanges), and context sentences varied by discourse condition: supportive (preceding sentences formed a coherent narrative; S) or unsupportive (preceding sentences unrelated; US) (**Table 2**). We also report comprehension data from a direct replication of [3] conducted on Prolific with N = 178 for comparison (Fig 1F).

Results: All GPT models exceeded 75% accuracy on fillers (standard inclusion criterion for studies with humans [1,3]), whereas LLama-3.3 attained 74.8%. Plausible sentences were interpreted literally at very high rates (>88%) across all runs for all models. Implausible sentences were more likely to be interpreted non-literally across all models (**Fig 1**). PO sentences were more likely to be interpreted literally than DO sentences (US: β = -1.48, 95% CrI = [-2.13, -.83]; S: β = -1.04, 95% CrI = [-1.66, -.42]). Active sentences had the highest rate of literal interpretation and were more likely to be interpreted literally than PO sentences (US: β = -1.22, 95% CrI = [-1.95, -.48]; S: β = -1.15, 95% CrI = [-1.82, -.48]), as were passives (US: β = -1.48, 95% CrI = [-2.54, -.49]; S: β = -1.94, 95% CrI = [-2.96, -.92]). Context effects were dependent on the structure: supportive context increased inference rates for PO ($\beta = 0.63$, 95% CrI = [.22, 1.05]) and active ($\beta = .7$, 95% CrI = [.18, 1.3]) but only slightly for DO ($\beta = .19$, 95% CrI = [-.14, .51]) and passive ($\beta = .18$, 95% CrI = [-.5, .87]). GPT-4o, GPT-4.1, and Llama-3.3 made more inferences than humans (see Fig 1F), while GPT-o3 and GPT-5 made fewer. Despite making fewer inferences overall, GPT-o3 and GPT-5 still made more inferences than humans on DO sentences.

Conclusions: Across all models, replicating previous findings with humans [1,3], DO sentences consistently elicited the most inferences, while PO, active, and passive sentences were interpreted more literally. Supportive discourse contexts increased the probability of inference relative to unsupportive contexts. These patterns suggest that, like humans, LLMs make noisy-channel inferences, which are sensitive to both the prior and noise likelihood. However, there was substantial variability across LLMs and they diverged from humans in the magnitude of structure effects, suggesting that the noise model (and the prior) may differ in LLMs relative to humans and between LLMs. Notably, GPT-o3 and GPT-5, which are designed to engage in deeper "reasoning" and be less susceptible to hallucinations, were less likely to make inferences (except for DO sentences).

References:

1. Gibson, E., Bergen, L., and Piantadosi, S. T. (2013). *PNAS, 110(20)*, 8051–8056. **2.** Levy, R. (2008). *Proceedings of the 13th Conference on Empirical Methods in NLP*, 234–243. **3.** Chen, S., Nathaniel, S., Ryskin, R., and Gibson, E. (2023). *Cognition, 238*, 105503. **4.** McCoy, R. T., Yao, S., Friedman, D., & Griffiths, T. L. (2023). *Processings of the National Academy of Sciences, 121(41)*, e2322420121. **5.** Cai, Z. G., Duan, X., Haslett, D. A., Wang, S., and Pickering, M. J. (2023). *arXiv*, 2303.08014.

Context Sentence 1	Context Sentence 2	Target Sentence	Comprehension Question
When the power outage happened, the daughter asked the mother for a candle.	The mother found a candle in the kitchen cabinet.	The mother gave the candle the daughter.	Did the daughter receive something/someone?

Table 1: Example of a Supportive Context trial.

Structure	Implausible Example	Possible Noise Types
Direct-Object (DO)	The mother gave the candle the daughter.	One-word deletion of “to”
Prepositional-Object (PO)	The mother gave the daughter to the candle.	One-word insertion of “to”
Active	The ball kicked the girl.	Two-word exchange of “girl” and “ball”
Passive	The girl was kicked by the ball.	Two-word exchange of “girl” and “ball”

Table 2: Examples of Implausible Sentences by Structure.

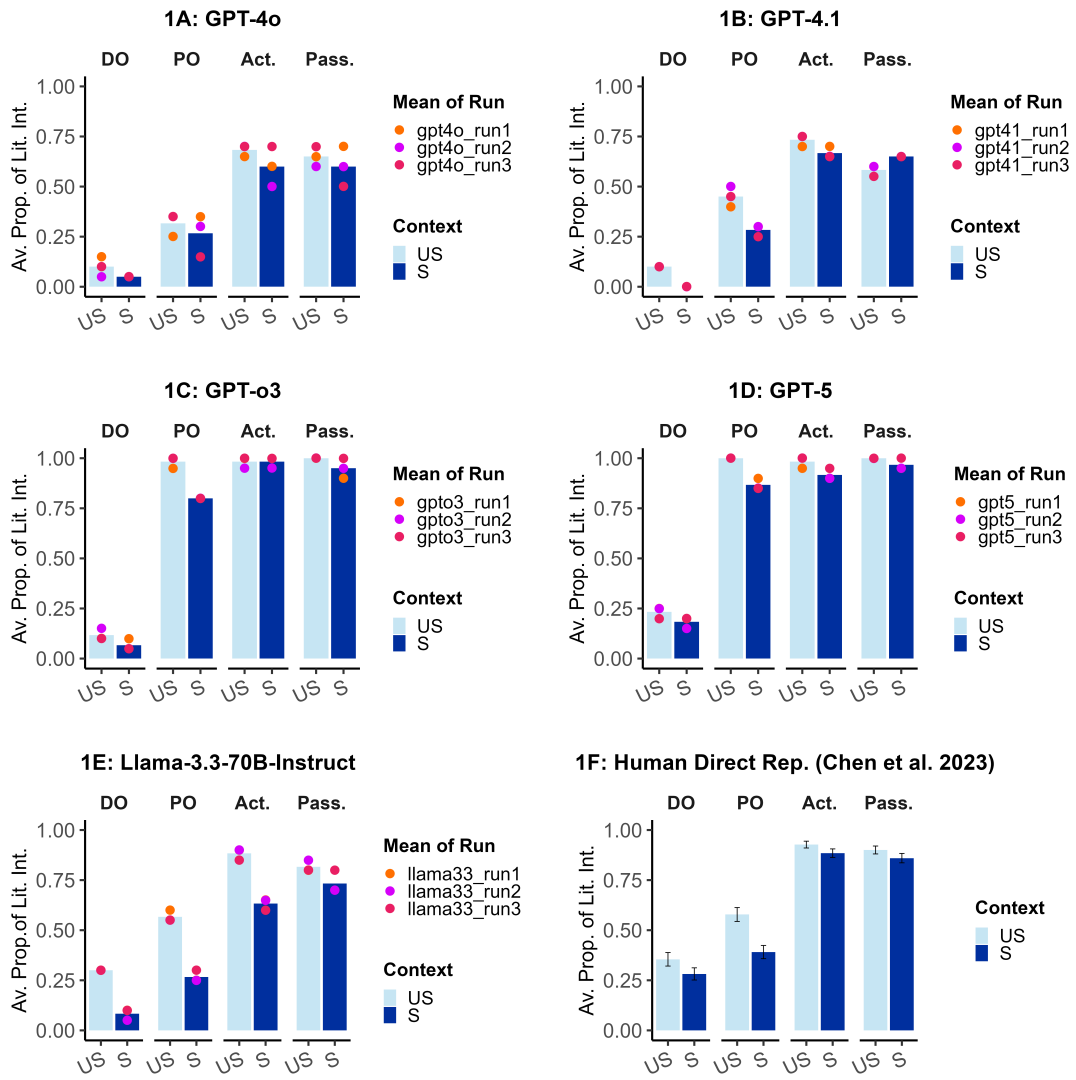

Figure 1: Proportions of literal interpretations for implausible sentences. S = Supportive discourse context. US = Unsupportive discourse context.